

A Miniaturised High-Pass Frequency Selective Surface for Sub-6 GHz Shielding Applications

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Abstract—This paper investigates an ultra-low-profile, high-pass frequency-selective surface (FSS) for sub-6-GHz shielding applications. The FSS exhibits high pass behaviour with the -10 dB threshold at 5 GHz with unit cell size of 4.86 mm. A 0.2 mm FR4 substrate provides mechanical stability to the single layer structure. A parametric study demonstrates tuning of the high-pass cut-off frequencies while maintaining the same unit-cell periodicity. The design was analysed in Ansys HFSS.

Index Terms—Frequency-Selective Surface, Electromagnetic Shielding, Ultra-low profile, Sub-6 GHz applications, High Pass filters.

I. INTRODUCTION

The sub-6 GHz band overlaps significantly with 5G NR Frequency Range 1 (FR1), which extends from 410 MHz to 7.125 GHz [1]. It is also utilised by multiple generations and classes of radio services, numerous cellular, satellite, navigation, and radar systems. The increasing spectrum congestion and proximity of devices utilising these bands raise significant concerns regarding electromagnetic interference. To mitigate unwanted frequency interactions, 2D or 3D structures called Frequency-selective surfaces (FSS) are employed. At low frequencies, application of FSS is constrained by the longer wavelengths which results in increased unit cell dimensions [2]. Therefore, several miniaturisation techniques have been investigated, such as increasing the path length of current by using convoluted, meandering or interleaved geometries, using high-permittivity substrates, incorporating capacitive or inductive loading [3]–[5]. Another method to achieve miniaturisation is to use a pair of an FSS and its complemented form. The increased electrical length aids in reducing the unit cell size [6]. This work designs an FSS whose resonant transmission features occur at much higher frequencies, then geometrically complements that same geometry so that its dual response produces the required high-pass transmission characteristics.

II. DESIGN

The complete structure, including its simulation environment, excitation ports, and boundary conditions, is illustrated in Fig. 1 (a)–(c), respectively. A hexagonal lattice can enable denser packing of unit cells, thereby delaying the onset of grating lobes. However, as a non-orthogonal lattice, it requires a more complex periodic simulation setup. A 0.2 mm thick

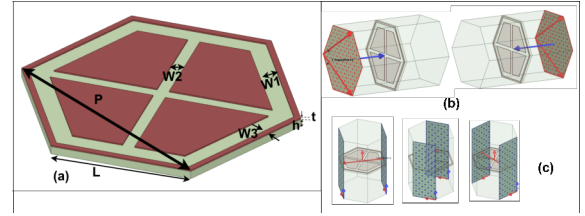


Fig. 1. (a) Proposed FSS in Ansys HFSS environment, (b) Floquet Port placement for a hexagonal unit cell, (c) Master-Slave Boundary placement for a hexagonal unit cell

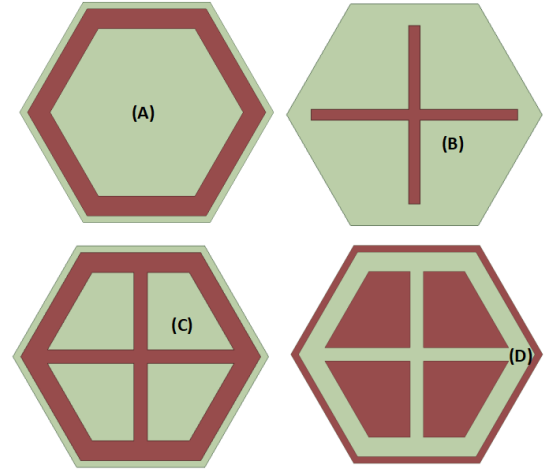


Fig. 2. The evolution of structure: (A) Stage 1: Hexagonal Loop, (B) Stage 2: Crossed dipole, (C) Stage 3: FSS, (D) Stage 4: complemented FSS

FR4 ($\epsilon_r=4.44$, $\tan\delta=0.02$) substrate is used. The design is monitored for its the S_{21} performance at higher frequencies, and the S_{11} performance at lower frequencies throughout its evolution up to third stage. After which, the focus reverses. All the stages are shown in Fig. 2, in the order of their development. The first stage comprises a hexagonal loop. When placed within the hexagonal lattice, it offers nearly uniform inter-cell capacitance in addition to its broadband response. To improve the characteristics of the S_{11} curve, a crossed dipole is introduced within the hexagonal loop. Although to conform with the C_6 rotational symmetry of the

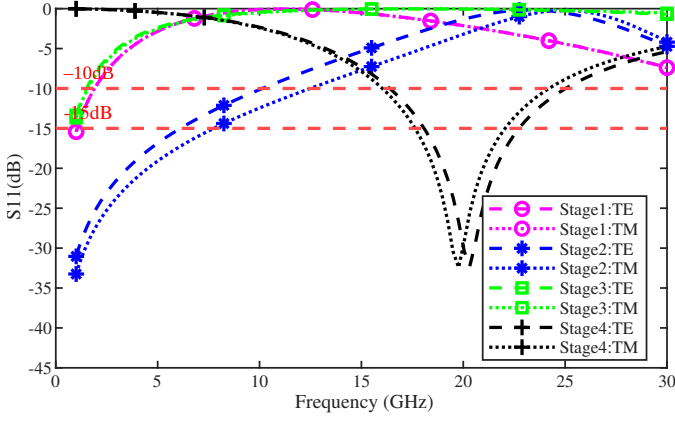


Fig. 3. The S_{11} (dB) curve for TE and TM for all the stages

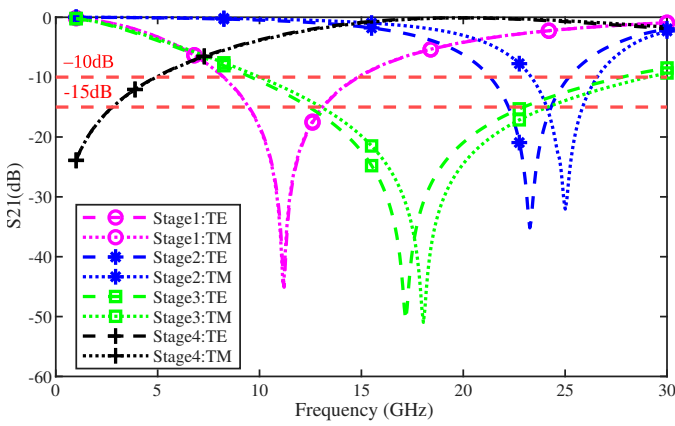


Fig. 4. The S_{21} (dB) curve for TE and TM for all the stages

hexagonal elements, a better choice would have been another C_6 element, a crossed dipole is selected to accommodate fabrication constraints. At a small scale, choosing a more complex structure could adversely affect fabrication feasibility. The two stages are combined to form the configuration shown in stage 3, which is then geometrically complemented to form the final configuration of Stage 4. The complementary transformation produces an approximate interchange of the transmission and reflection characteristics [6], [7]. The values of the physical parameters are provided in Table I.

TABLE I
PHYSICAL DIMENSIONS OF THE PROPOSED STRUCTURE SHOWN IN FIG. 1.

S.No.	Parameter	Value (mm)
1	L	2.7
2	P	4.86
3	W_1	0.5
4	W_2	0.3
5	W_3	0.15
6	t	0.035
7	h	0.2

III. RESULTS

The FSS was simulated using Ansys High-Frequency Structure Simulation software. The results are shown in Fig. 3 and Fig. 4. The results are analysed at two thresholds, -10 dB and -15 dB, to provide a clearer assessment of performance, as the transition from stopband to passband occurs gradually. For stage 1, the transmission characteristics exhibit a -10 dB bandwidth of 6.6 GHz with a -45 dB dip at 11.2 GHz. The -15 dB bandwidth is 3.6 GHz. The transverse electric (TE) and transverse magnetic (TM) modes coincide, indicating polarisation insensitivity consistent with the C_6 rotational symmetry of the geometry. The reflection characteristics indicate a -10 dB threshold at 1 GHz and a -15 dB threshold at 1.9 GHz. In stage 2, the transmission curve for the TE mode shows a -10 dB bandwidth of 3.9 GHz, with a -35 dB dip at 23.3 GHz, and a -15 dB bandwidth of 2 GHz. The TM mode exhibits a -10 dB bandwidth of 3.2 GHz with a -25 dB dip at 25 GHz, and a -15 dB bandwidth of 1.8 GHz. The S_{11} curve indicates the -10 dB threshold at 10.1 GHz and the -15 dB threshold at 6.1 GHz for TE, and the -10 dB threshold at 12.5 GHz and the -15 dB threshold at 7.8 GHz for TM. The polarisations are separated by 1.7 GHz. In stage 3, both structural elements are combined. The difference between TE and TM modes for the S_{21} curve is reduced to 1 GHz. For TE, the S_{21} curve shows a -10 dB bandwidth of 13 GHz with a -50 dB dip at 17 GHz, and a -15 dB bandwidth of 10 GHz. For TM, the -10 dB bandwidth is 19 GHz with a -51 dB dip at 18 GHz, and the -15 dB bandwidth is 11 GHz. The S_{11} curve indicates the -10 dB threshold at 1.51 GHz for TE and 1.57 GHz for TM. A clear resemblance is observed between the reflection characteristics of stage 3 and the transmission characteristics of the stage 4 response. The S_{21} now exhibits high-pass behaviour. The -10 dB threshold for the TE mode occurs at 4.94 GHz, and the -15 dB threshold at 2.79 GHz. For the TM mode, the -10 dB threshold is at 4.9 GHz and the -15 dB threshold at 2.77 GHz. The difference between the two modes for the transmission curve is now only a few tens of MHz.

The width of the hexagonal slot can be used to control the frequencies at which the transmission curves cross the -10 dB and -15 dB thresholds without changing the unit-cell periodicity. Fig. 5 illustrates the variation of the TE mode of S_{21} with W_1 . Since the focus is on the sub-6 GHz range, the simulation's frequency range is reduced for better clarity. The graph shows that a reduction in W_1 shifts the curve toward higher frequencies and steepens the transition from stopband to passband. Fig. 6 shows that the TM mode follows a similar pattern, except there is not much effect on the transition.

IV. CONCLUSION AND FUTURE SCOPE

This study investigates an ultra-low-profile FSS, shielding up to 2.77 GHz with $S_{21} < -15$ dB and up to 5 GHz with $S_{21} < -10$ dB. A comparison between previously reported papers and this work is given in Table II. A direct comparison between bandstop and high-pass structures is challenging. To establish a common basis for comparison, P/λ_0 is calculated at the lowest resonant frequency, f_L , reported in each

TABLE II
COMPARISON OF THE PROPOSED FSS WITH MINIATURISED FSS DESIGNS

Ref. No.	f_L (GHz)	Substrate	Thickness, ϵ_r , $\tan\delta$	Unit-cell size	FSS Layers	P/λ_0 of this work at f_L
[8]	0.90	Not Reported	1.6 mm, $\epsilon_r = 4.4$, $\tan\delta = 0.02$	$0.078\lambda_0$	2	0.0146
[9]	1.056	FR4	1.6 mm, $\epsilon_r = 4.3$, $\tan\delta = 0.025$	$0.034\lambda_0$	2	0.0171
[10]	1.10	FR4	1.6 mm, $\epsilon_r = 4.4$, $\tan\delta = 0.023$	$0.022\lambda_0$	3	0.0178
[11]	4	FR4	0.25 mm, $\epsilon_r = 4.4$, $\tan\delta = 0.02$	$0.091\lambda_0$	1	0.0648
[12]	1.80	Soda lime glass	2 mm, $\epsilon_r = 6.5$, $\tan\delta = 0.007$	$0.111\lambda_0$	1	0.0292
[13]	2.40	FR-4	0.8 mm, $\epsilon_r = 4.3$, $\tan\delta = 0.025$	$0.050\lambda_0$	1	0.0389
[14]	2.10	Glass	2 mm and 1.7 mm, $\epsilon_r = 6.45$ and 6.15	$0.0714\lambda_0$	1	0.0340

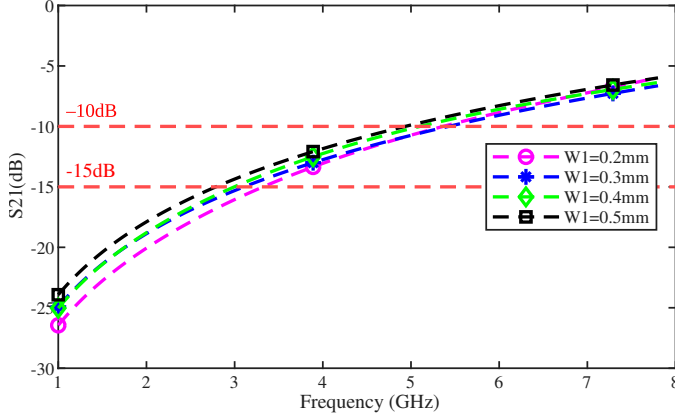


Fig. 5. The S_{21} (dB) curves for different $W1$ values: TE

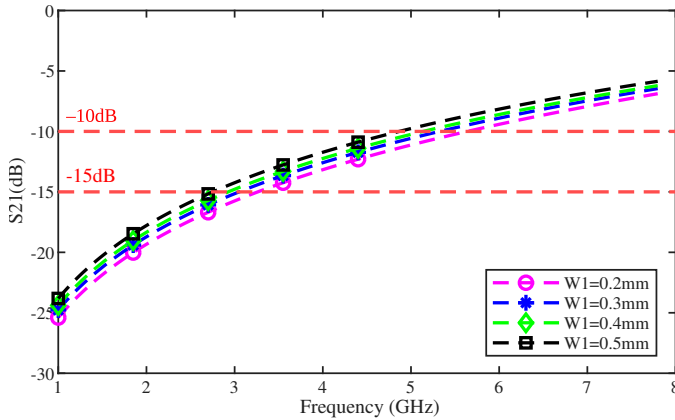


Fig. 6. The S_{21} (dB) curves for different $W1$ values: TM

referenced study. The transmission response can be tuned by changing the dimensions. Experimental validation of the simulation results and a study on stability at oblique incidence will be addressed in future work. The transmission characteristics of the proposed configuration exhibit polarisation insensitivity at normal incidence since the TE and TM modes differ by only 20–40 MHz. In its current form, the proposed FSS is potentially suitable for low-profile shielding panels and electronic enclosures requiring suppression of sub-6-GHz electromagnetic interference.

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